

SI 140A: Probability and Statistics for EECS

Spring 2025, Instructor: Xavier Lagorce, TA: Junye Wang, Chuyan Zhou

Final, 2025/06/12, 10:30 am – 12:30 pm

Final Exam

Time Limit: 120 Minutes

Name: _____

Student ID: _____

This exam contains 12 pages (including this cover page) and 8 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

Try to answer as many problems as you can. The following rules apply:

- All examination rules must be strictly observed throughout the entire test, and any form of cheating is prohibited.
- Other than allowable materials, students taking closed-book tests must place their books, notes, tablets, and any other electronic devices in places designated by the examiners.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.
- Please try to simplify your results as much as possible. An answer that could have been further simplified will receive partial credits.

Do not write in the table below.

Problem	1	2	3	4	5	6	7	8
Points	10	10	10	10	10	15	15	20
Score								

This page is left blank intentionally.

Problem 1

Basic Concepts.

- (a) (5') Please describe the differences and connections between Probability and Statistics.
- (b) (5') Please describe the pros and cons of Bayesian and frequentist statistical inference. Then explain why conjugate priors are important for Bayesian statistical inference.

Problem 2

Let $X \sim \text{Beta}(3, 2)$. Consider its PDF, and

- (a) (3') find the CDF of X .
- (b) (3') find $P(0 < X < \frac{1}{2})$.
- (c) (4') Find the mean and variance of X (without quoting results about the Beta distribution, but you can use that to verify your results).

Problem 3

Let X and Y be two continuous r.v.s with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6xy, & \text{if } 0 \leq x \leq 1, 0 \leq y \leq \sqrt{x}, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) (5') Find the marginal distributions of X and Y . Are X and Y independent?
- (b) (5') Find $E[X|Y = y]$ and $\text{Var}[X|Y = y]$ for $0 \leq y \leq 1$.

Problem 4

Let U_1, U_2, U_3 be i.i.d. $\text{Unif}(0, 1)$, and let $L = \min(U_1, U_2, U_3)$, $M = \max(U_1, U_2, U_3)$.

- (a) (5') Find the marginal CDF and marginal PDF of M , and the joint CDF and joint PDF of L, M . Hint: For the latter, start by considering $P(L \geq l, M \leq m)$.
- (b) (5') Find the conditional PDF of M given L .

Problem 5

Laplace's law of succession says that if $X_1, X_2, \dots, X_{n+1}$ are conditionally independent Bern(p) r.v.s given p , but p is given a Unif(0, 1) prior to reflect ignorance about its value, then

$$P(X_{n+1} = 1 | X_1 + \dots + X_n = k) = \frac{k+1}{n+2}.$$

As an example, Laplace discussed the problem of predicting whether the sun will rise tomorrow, given that the sun did rise every time for all n days of recorded history; the above formula then gives $(n+1)/(n+2)$ as the probability of the sun rising tomorrow (of course, assuming independent trials with p unchanging over time may be a very unreasonable model for the sunrise problem).

- (a) (5') Find the posterior distribution of p given $X_1 = x_1, \dots, X_n = x_n$, and show that it only depends on the sum of the $x_j, j \in \{1, \dots, n\}$.
- (b) (5') Prove Laplace's law of succession, using a form of LOTP to find

$$P(X_{n+1} = 1 | X_1 + \dots + X_n = k)$$

by conditioning on p .

Problem 6

We wish to estimate an unknown parameter θ , based on an r.v. X we will get to observe. As in the Bayesian perspective, assume that X and θ have a joint distribution. Let $\hat{\theta}$ be the estimator (which is a function of X). Then $\hat{\theta}$ is said to be *unbiased* if $E(\hat{\theta}|\theta) = \theta$, and $\hat{\theta}$ is said to be the *Bayes procedure* if $E(\theta|X) = \hat{\theta}$.

- (a) (5') Let $\hat{\theta}$ be unbiased. Find $E[(\hat{\theta} - \theta)^2]$ (the average squared difference between the estimator and the true value of θ), in terms of marginal moments of $\hat{\theta}$ and θ . Hint: **Condition on θ .**
- (b) (5') Repeat (a), except in this part suppose that $\hat{\theta}$ is the *Bayes procedure* rather than assuming that it is unbiased. Hint: **Condition on X .**
- (c) (5') Show that it is **impossible** for $\hat{\theta}$ to be both the Bayes procedure and unbiased, except in silly problems where we get to know θ perfectly by observing X . Hint: If Y is a nonnegative r.v. with mean 0, then $P(Y = 0) = 1$.

Problem 7

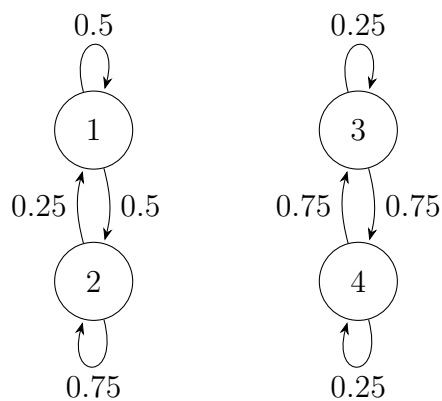
A DNA sequence can be represented as a sequence of letters, where the “alphabet” has 4 letters: A,C,T,G. Suppose such a sequence is generated randomly, where the letters are independent and the probabilities of A,C,T,G are p_1, p_2, p_3, p_4 respectively.

- (a) (5') In a DNA sequence of length 115, what is the expected number of occurrences of the expression “CATCAT” in terms of the p_j ? Note that, for example, the expression “CATCATCAT” counts as 2 occurrences.
- (b) (5') What is the probability that the first A appears earlier than the first C appears, as letters are generated one by one (in terms of the p_j)? Hint: **Consider X as a random variable represent the first letter which is A or C.**
- (c) (5') For this part, assume that the p_j are unknown. Suppose we treat p_2 as a $\text{Unif}(0, 1)$ r.v. before observing any data, and that then the first 3 letters observed are “CAT”. Given the information we know about X as the number of C's in the data, what is the probability that the next letter is C? Hint: **You may sanity check**

$$P(S_4 = C|\text{data}) = \int_0^1 P(S_4 = C|p_2 = p)f_{p_2}(p|\text{data})dp.$$

Problem 8

Consider the Markov chain shown below, with state space $\{1, 2, 3, 4\}$ and the labels on the arrows indicate transition probabilities.



- (a) (5') Write down the transition matrix Q for this chain. Which states (if any) are **recurrent**? Which states (if any) are **transient**?
- (b) (5') There are two different stationary distributions for the chain. Find them.

A cat and a mouse move independently back and forth between two rooms. At each time step, the cat moves from the current room to the other room with probability 0.8. Starting from room 1, the mouse moves to room 2 with probability 0.3 (and remains otherwise). Starting from room 2, the mouse moves to room 1 with probability 0.6 (and remains otherwise).

- (c) (5') Find the stationary distributions of the cat chain and of the mouse chain.

Note that there are 4 possible (cat, mouse) states: both in room 1, cat in room 1 and mouse in room 2, cat in room 2 and mouse in room 1, and both in room 2. Number these cases 1, 2, 3, 4, respectively, and let Z_n be the number representing the (cat, mouse) state at time n .

Is $Z_0, Z_1, Z_2, \dots$ a Markov chain?

- (d) (5') Now suppose that the cat will eat the mouse if they are in the same room. We wish to know the expected time (number of steps taken) until the cat eats the mouse for two initial configurations: when the cat starts in room 1 and the mouse starts in room 2, and vice versa. Set up a system of two linear equations in two unknowns whose solution is the desired values and solve it.

Y discrete		Y continuous	
X discrete	$P(Y = y X = x) = \frac{P(X=x Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y X = x) = \frac{P(X=x Y=y)f_Y(y)}{P(X=x)}$	
X continuous	$P(Y = y X = x) = \frac{f_X(x Y=y)P(Y=y)}{f_X(x)}$	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}$	

Y discrete		Y continuous	
X discrete	$P(X = x) = \sum_y P(X = x Y = y)P(Y = y)$	$P(X = x) = \int_{-\infty}^{\infty} P(X = x Y = y)f_Y(y)dy$	
X continuous	$f_X(x) = \sum_y f_X(x Y = y)P(Y = y)$	$f_X(x) = \int_{-\infty}^{\infty} f_{X Y}(x y)f_Y(y)dy$	

Figure 1: The Bayes' Rule and LOTP.

Name	Param.	PMF or PDF	Mean	Variance
Bernoulli	p	$P(X = 1) = p, P(X = 0) = q$	p	pq
Binomial	n, p	$\binom{n}{k}p^kq^{n-k}$, for $k \in \{0, 1, \dots, n\}$	np	npq
FS	p	pq^{k-1} , for $k \in \{1, 2, \dots\}$	$1/p$	q/p^2
Geom	p	pq^k , for $k \in \{0, 1, 2, \dots\}$	q/p	q/p^2
NBinom	r, p	$\binom{r+n-1}{r-1}p^rq^n$, $n \in \{0, 1, 2, \dots\}$	rq/p	rq/p^2
HGeom	w, b, n	$\frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}}$, for $k \in \{0, 1, \dots, n\}$	$\mu = \frac{nw}{w+b}$	$\left(\frac{w+b-n}{w+b-1}\right)n\frac{\mu}{n}(1 - \frac{\mu}{n})$
Poisson	λ	$\frac{e^{-\lambda}\lambda^k}{k!}$, for $k \in \{0, 1, 2, \dots\}$	λ	λ
Uniform	$a < b$	$\frac{1}{b-a}$, for $x \in (a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	μ, σ^2	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2
Log-Normal	μ, σ^2	$\frac{1}{x\sigma\sqrt{2\pi}}e^{-(\log x - \mu)^2/(2\sigma^2)}$, $x > 0$	$\theta = e^{\mu+\sigma^2/2}$	$\theta^2(e^{\sigma^2} - 1)$
Expo	λ	$\lambda e^{-\lambda x}$, for $x > 0$	$1/\lambda$	$1/\lambda^2$
Gamma	a, λ	$\Gamma(a)^{-1}(\lambda x)^a e^{-\lambda x} x^{a-1}$, for $x > 0$	a/λ	a/λ^2
Beta	a, b	$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}x^{a-1}(1-x)^{b-1}$, for $0 < x < 1$	$\mu = \frac{a}{a+b}$	$\frac{\mu(1-\mu)}{a+b+1}$
Chi-Square	n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$, for $x > 0$	n	$2n$
Student- t	n	$\frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\Gamma(n/2)}(1+x^2/n)^{-(n+1)/2}$	0 if $n > 1$	$\frac{n}{n-2}$ if $n > 2$

Figure 2: Some properties of distributions.